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 Studierichting: Informatica
 Oefeningenreeks 3G nr.11.1

$$f(x) = \frac{2x}{x^2 + 1}$$

$$\text{V.A: } \lim_{x \rightarrow 0} \frac{2x}{x^2 + 1} = 0 \text{ geen V.A}$$

$$\text{H.A: } \lim_{x \rightarrow \infty} \frac{2x}{x^2 + 1} = 0 \text{ T.O}$$

$$\begin{array}{c|c} x & 0 \\ \hline f(x) & - \quad 0 \quad + \end{array}$$

Dus:

Als $x \rightarrow +\infty$ dan is de limiet 0^+

Als $x \rightarrow -\infty$ dan is de limiet 0^-

$$\begin{aligned} f'(x) &= 2 \cdot \left(\frac{x}{x^2 + 1} \right)' \\ &= 2 \cdot \frac{(x^2 + 1) \cdot 1 - 2x^2}{(x^2 + 1)^2} \\ &= 2 \cdot \frac{-x^2 + 1}{(x^2 + 1)^2} = \frac{-2(x^2 - 1)}{(x^2 + 1)^2} \\ f''(x) &= -2 \cdot \left(\frac{x^2 - 1}{(x^2 + 1)^2} \right)' \\ &= -2 \cdot \frac{(x^2 + 1)^2 \cdot 2x - 4x(x^2 + 1)(x^2 - 1)}{(x^2 + 1)^4} \\ &= -2 \cdot \frac{(x^2 + 1) \cdot (2x(x^2 + 1) - 4x(x^2 - 1))}{(x^2 + 1)^4} \\ &= -2 \cdot \frac{(2x(x^2 + 1 - 2x^2 + 2))}{(x^2 + 1)^3} \\ &= \frac{4x(x^2 - 3)}{(x^2 + 1)^3} \end{aligned}$$

Nulpunten van:

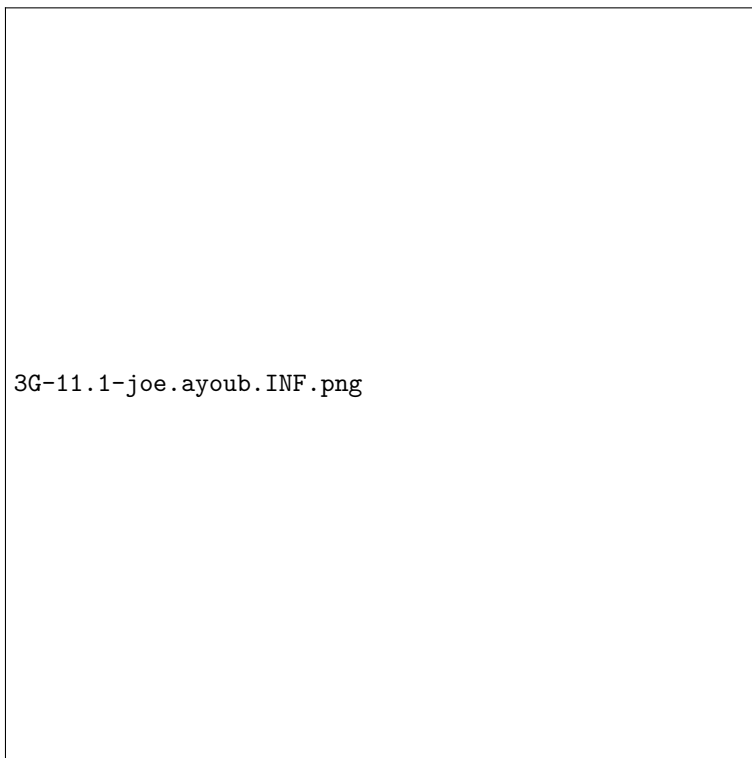
- $f(x) = \{0\}$
- $f'(x) = \{-1, 1\}$
- $f''(x) = \{-\sqrt{3}, 0, \sqrt{3}\}$

x	$-\sqrt{3}$	-1	0	1	$\sqrt{3}$
$f(x)$	-	-	-	+	+
$f'(x)$	-	-	0	+	0
$f''(x)$	-	0	+	+	0
$f(x)$	$\searrow$ (	$\searrow$ B	$\searrow$ (	$m(1)$ $\nearrow$ (	$\nearrow$ B

Coördinaten van de buigpunten:

- $\left(-\sqrt{3}, \frac{-\sqrt{3}}{2}\right)$
- $(0, 0)$
- $\left(\sqrt{3}, \frac{\sqrt{3}}{2}\right)$

Tekening:



References